Fuel Gauge

R2 = (R1 * V_OUT) / (V_IN − V_OUT)
R2 = (300k * 0.93V) / (12.6V − 0.93V) = 23.91kΩ

V_OUT = V_IN * (R2 / (R1 + R2))
V_OUT = 12.6V * (24.9k / (300k + 24.9k)) = 0.966V

AFE_C3
R3 100k
Q1 ZXMP10A13FTA
R6 100k
Q2 BSS123
GATE_DRIVER_PMON_EN
R2 300k±0.1% 25ppm/°C
C1 3n3
R5 24k9±0.1% 25ppm/°C
GND
GATE_DRIVER_PACKDIV
R1 300k±0.1% 25ppm/°C
R4 300k±0.1% 25ppm/°C
C2 3n3
R7 25ppm/°C 10k±0.1%
GND
AFE_REGOUT
R9 100k
R10 100k
C3 4u7
C4 1u
C5 100n
GND
U1 BQ78350DBT-R1A
MRST
BAT
VEN
VAUX
SCL
SDA
ALERT
PRECHG
PRES
KEYIN
SAFE
GPIO_A
GPIO_B
VCC
VSS
LED1 YELGREEN D1
LED2 YELGREEN D2
LED3 YELGREEN D3
LED4 YELGREEN D4
LED5 YELGREEN D5
COM
DISP
PWRM
SMBD
SMBC
ADREN
SMBA
RBI
addr: 0x16
C6 100n
GND
J2 AFE_SCL AFE_SDA AFE_ALERT GND BMS_I2C
R8 510k
GATE_DRIVER_PCHG_EN
R11 100R
R12 100R
From slvu925c
R13 1M
R14 1M
D6 5V6
D7 5V6
from TIDA-00792
Q3 MMBT3904
AFE_C3
AFE_CAP1
C7 1u
C8 100n
GND
R17 220k
S1
BATT
GND
R18 510k
GATE_DRIVER_CP_EN
R15 100R
R16 100R
D8
D9
R19 10k
R20 10k
J3
J4
MCU_P3V3
U2 PCA9306D,118
VREF1
VREF2
EN
SCL1
SDA1
SCL2
SDA2
GND
MCU_SMBC
MCU_SMBD
3V3
SMB
EV2300-002
NC (4)
SMBD (3)
SMBC (2)
GND (1)
J7
SMBD_2V5
SMBC_2V5
GND
J1
GND
AFE_SDA
AFE_C3
AFE_ALERT
AFE_CAP1
AFE_SCL
AFE_REGOUT
GND
6120XX21621_61200821621
J5
GND
GATE_DRIVER_PMON_EN
GATE_DRIVER_CP_EN
GATE_DRIVER_PCHG_EN
GATE_DRIVER_PACKDIV
GND
6120XX21621_61200621621
J6
GND
MCU_P3V3
MCU_SMBC
MCU_SMBD
GND
6120XX21621_61200621621

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